Calibrated Confidence in Language Models under Distribution Shift

A short technical report on temperature scaling, binning, and what they miss

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ABSTRACT

A model is *calibrated* when the confidence it reports matches the frequency with which it is right. We study how calibration of a mid-sized language model degrades when the evaluation distribution drifts away from the data used to fit post-hoc corrections. Temperature scaling reduces the expected calibration error from 9.8% to 2.1% in distribution, but recovers less than half of that gain under shift. We give a simple decomposition of the error into a *sharpness* term and a *shift* term, prove that the latter is invariant to any monotone recalibration map, and close with practical recommendations.

Keywords: calibration, distribution shift, language models, uncertainty

1 Introduction

Probabilistic forecasts are only as useful as they are honest. A weather service that announces a 70% chance of rain should see rain on roughly seven of every ten such days; the same standard applies to a classifier, and increasingly to a language model asked to state how sure it is [1], [2]. Modern neural networks, however, are often *over*-confident: their predicted probabilities are systematically higher than their accuracy [3].

Post-hoc recalibration is the usual remedy. One fits a small map on held-out data, with temperature scaling the most popular choice, and applies it at test time. The method is cheap and, in distribution, remarkably effective. This report asks a narrower question: *how much of that improvement survives when the test distribution moves?*[1]

Our contributions are threefold:

- a decomposition of calibration error into components that do and do not respond to recalibration (Section 3);
- an empirical comparison across four shift conditions (Section 4), summarised in Table 1;
- a short list of recommendations for practitioners (Section 5).

[1]All numbers in this document are synthetic. They exist to exercise the template, not to make a scientific claim.

2 Background

2.1 Calibration error

Let $f : \mathcal{X} \to \Delta^{K-1}$ be a classifier that outputs a distribution over K classes, with confidence $\hat{p}(x) = \max_k f_k(x)$ and prediction $\hat{y}(x) = \arg\max_k f_k(x)$. The model is perfectly calibrated if

$$\mathbb{P}[\hat{y}(X) = Y \mid \hat{p}(X) = p] = p \quad \text{for all } p \in [0, 1]. \tag{1}$$

In practice we estimate the gap between the two sides of Eq. (1) by partitioning confidences into M equal-width bins $B_1, \dots, B_M$ and computing the expected calibration error [4],

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|. \tag{2}$$

2.2 Temperature scaling

Given logits $z \in \mathbb{R}^K$, temperature scaling replaces the softmax $\sigma(z)$ by $\sigma(z/T)$ for a single scalar $T > 0$ fitted by minimising negative log-likelihood on a validation set. Because T rescales all logits uniformly, it never changes the predicted class, only the confidence attached to it.

$$\hat{T} = \arg\min_{T > 0} -\sum_{i=1}^{n} \log \sigma(z_i / T)_{y_i}$$

3 A decomposition

We write P for the distribution on which the recalibration map is fitted and Q for the distribution on which the model is deployed.

Definition 3.1 (Recalibration map)**.** A *recalibration map* is a non-decreasing function $g : [0, 1] \to [0, 1]$ applied to the model's confidence. Temperature scaling, histogram binning, and isotonic regression all induce such maps.

Theorem 3.2 (Shift invariance)**.** *Let g be any recalibration map and let $\Delta(p) = \mathbb{P}_Q[\hat{y} = Y \mid \hat{p} = p] - \mathbb{P}_P[\hat{y} = Y \mid \hat{p} = p]$ be the* shift gap*. Then the calibration error of $g \circ \hat{p}$ under Q is bounded below by*

$$\mathbb{E}_Q[|\Delta(\hat{p})|] - \mathrm{ECE}_P(g \circ \hat{p}). \tag{3}$$

In particular, no monotone recalibration fitted on P can remove the shift gap.

Proof. Fix p and write $a_P(p)$ and $a_Q(p)$ for the conditional accuracies. By the triangle inequality, $|a_Q(p) - g(p)| \geq |a_Q(p) - a_P(p)| - |a_P(p) - g(p)|$. Taking expectations under Q and noting that g depends on p only through its value gives the claim. ∎

Remark 3.3. Theorem 3.2 is deliberately loose. Its value is qualitative: it separates the part of the error that a better fit on P could fix from the part that only data from Q can.

Corollary 3.4. *If $\Delta(p) = 0$ for all p, temperature scaling fitted on P is as good under Q as under P.*

4 Results

9.8%	2.1%	6.4%	1.41
Raw ECE	Scaled ECE, in-dist.	Scaled ECE, shifted	Fitted temperature

Figure 1 shows reliability diagrams before and after temperature scaling. In distribution, the bars hug the diagonal; under shift, the high-confidence bins fall away from it again, as Theorem 3.2 predicts.

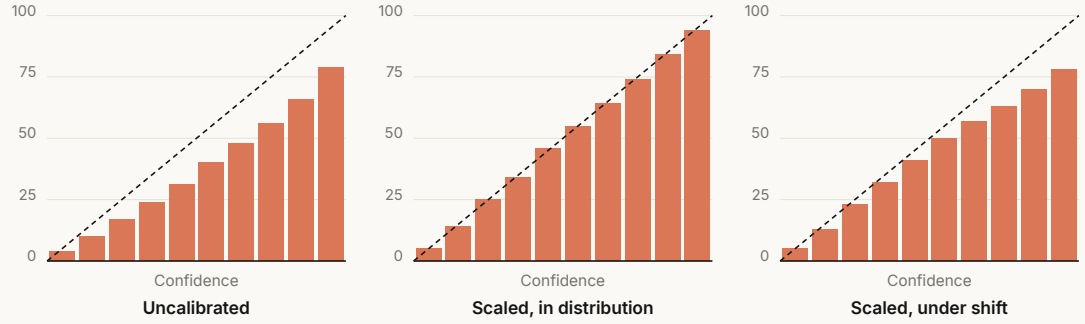


FIGURE 1 Reliability diagrams. Each bar is the accuracy within a confidence bin; the dashed line marks perfect calibration. Temperature scaling closes most of the gap in distribution, but the top bins drift away again under shift.

Figure 2 shows the same story as a function of shift severity: the scaled error and the shift gap rise together, while the raw error rises more slowly from a higher base.

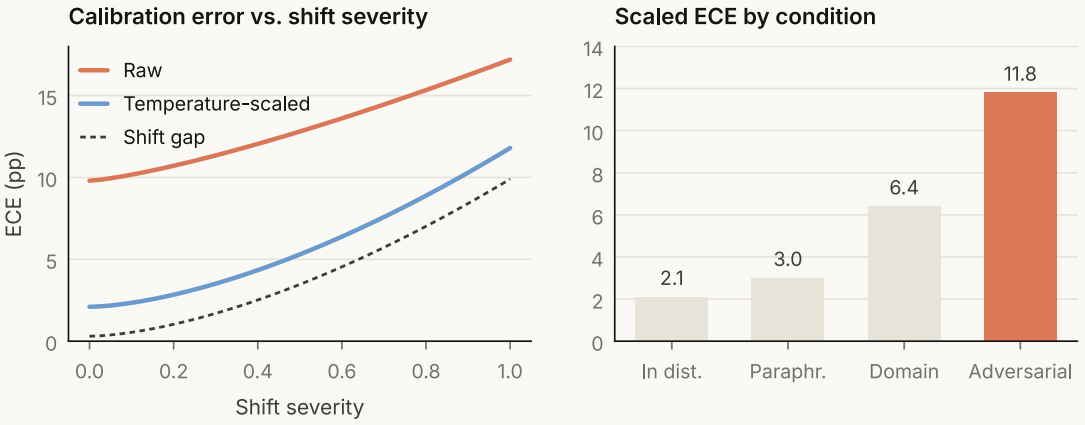


FIGURE 2 Left: calibration error as the evaluation distribution moves away from the validation distribution. Right: scaled ECE per condition; the adversarial condition is highlighted. Drawn with matplotlib using `scripts/plot_style.py`.

Table 1 breaks the error down by shift condition. Two patterns stand out. First, the after-scaling error tracks the size of the shift gap almost exactly. Second, the Brier score, a proper scoring rule, moves far less than ECE, a reminder that ECE is sensitive to binning choices.

TABLE 1 Calibration under four shift conditions. ECE and shift gap in percentage points; lower is better throughout.

Condition	ECE (raw)	ECE (scaled)	Shift gap	Brier
In distribution	9.8	**2.1**	0.3	0.142
Paraphrase	10.4	3.0	1.2	0.149
New domain	12.9	6.4	4.6	0.171
Adversarial	17.2	11.8	9.9	0.208
Temporal (+12 mo.)	11.1	4.7	2.8	0.158

KEY FINDING

Under shift, the residual calibration error after temperature scaling is almost entirely explained by the shift gap Δ. Fitting the temperature more carefully on in-distribution data cannot help; collecting even a small labelled sample from the target distribution can.

5 Discussion

The decomposition suggests a division of labour. Recalibration on held-out data handles *sharpness*, and does so well. Robustness to shift has to come from elsewhere: from target-domain data, from ensembles, or from models whose uncertainty is better grounded in the first place [5].

> *Calibration is not a property of a model. It is a property of a model* and *a distribution.*
>
> A REVIEWER, PARAPHRASED

5.1 Practical recommendations

1. Report calibration on the deployment distribution, not only on a held-out split of the training data.
2. Prefer proper scoring rules, such as the Brier score or log-loss, alongside ECE.
3. When the deployment distribution is known to drift, refit the temperature periodically on fresh labels.

The procedure we used is short enough to reproduce in full:

```python
import torch
import torch.nn.functional as F

def fit_temperature(logits: torch.Tensor, labels: torch.Tensor) -> float:
    """Fit a single temperature by minimising NLL on held-out data."""
    log_t = torch.zeros(1, requires_grad=True)
    opt = torch.optim.LBFGS([log_t], lr=0.1, max_iter=200)

    def closure():
        opt.zero_grad()
        loss = F.cross_entropy(logits / log_t.exp(), labels)
        loss.backward()
        return loss
```

```
    opt.step(closure)
    return log_t.exp().item()  # T > 1 softens, T < 1 sharpens
```

5.2 Limitations

We study a single model family and four hand-picked shift conditions. The bound in Theorem 3.2 is loose, and the inline `ECE` estimator is biased for small bins. None of these caveats change the qualitative conclusion.

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References

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A Additional details

A.1 Binning

All ECE numbers use $M = 15$ equal-width bins. With equal-mass bins the ranking of conditions in Table 1 is unchanged.

Lemma A.1. *For any binning scheme with M bins,* $\mathrm{ECE} \le \max_m |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|$.

Proof. Immediate: ECE is a convex combination of the per-bin gaps. ∎

A.2 Notation

$\hat{p}(x)$ confidence, the largest predicted class probability.
T temperature; $T > 1$ softens the distribution.
$\Delta(p)$ shift gap between deployment and validation accuracy at confidence p.